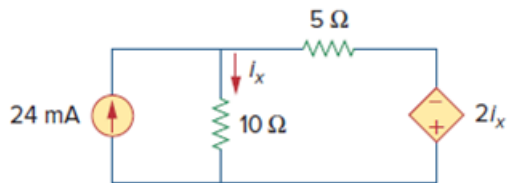


Problem

Use source transformation to find i_x in the circuit shown in Fig. 4.22.

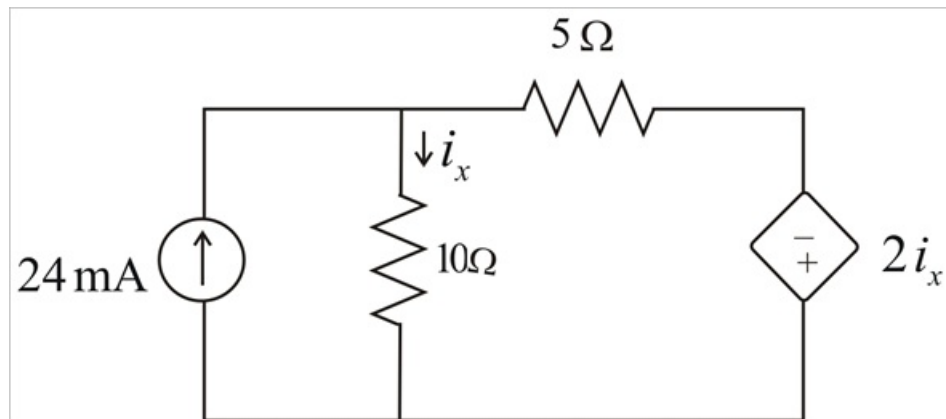
Figure. 4.22:



Step-by-step solution

Step 1 of 5

Given the circuit is



Step 2 of 5

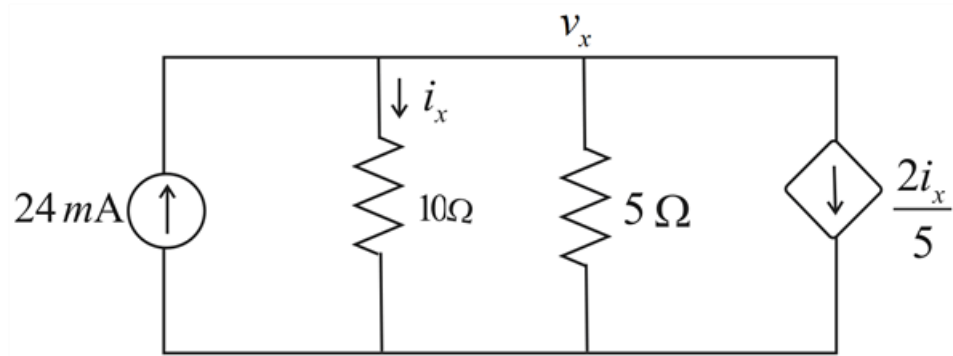
By source transformation we have the process of replacing a voltage source V_s in series with a resistor 'R' by a current source is in parallel with a resistor 'R' or vice versa

From the circuit dependent voltage source can be replaced by dependent current source

$$i_A = \frac{2i_x}{5}$$

Then redrawn circuit is

Step 3 of 5



Step 4 of 5

By KCL from the above circuit we have

$$-24 \times 10^{-3} + \frac{V_x}{10} + \frac{V_x}{5} + \frac{2i_x}{5} = 0$$

$$i_x = \frac{V_x}{10}$$

$$-24 \times 10^{-3} + \frac{V_x}{10} + \frac{V_x}{5} + \frac{2}{5} \times \frac{V_x}{10} = 0$$

$$-24 \times 10^{-3} + V_x \left[\frac{1}{10} + \frac{1}{5} + \frac{1}{25} \right] = 0$$

$$-24 \times 10^{-3} + 0.34 V_x = 0$$

$$V_x = \frac{24 \times 10^{-3}}{0.34}$$

$$V_x = 70.588 \times 10^{-3} \text{ V}$$

Step 5 of 5

We have

$$i_x = \frac{V_x}{10}$$

$$i_x = \frac{70.588 \times 10^{-3}}{10}$$

$$\boxed{i_x = 7.058 \text{ mA}}$$